

ADVANCED FUNCTIONAL MATERIALS

Supporting Information

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Machine Learning for Screening Small Molecules as Passivation Materials for Enhanced Perovskite Solar Cells

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Supporting Information

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Experimental Section

Machine-learning details

Pearson's correlation coefficient is used to eliminate redundant features. It is based on the covariance matrix of the data to evaluate the strength of relationship between two features. The detailed computational formula is as follow:

$$p = \frac{\sum_{i=1}^n [(x_i - \bar{x}) \times (y_i - \bar{y})]}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \times \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} \quad (1)$$

Where x_i and y_i are the values of two features in the i -th data ($i = 1, 2, 3 \dots, n$), $\bar{x}$ and $\bar{y}$ are the average values of these two features in n data. When the absolute value of p is larger than 0.8, it indicates a strong correlation between two features. In such instances, we can opt for a single representative feature and discard the other to reduce redundancy.

To enhance data continuity, it is essential to normalize all input X data. This normalization process is accomplished using the *StandardScaler* algorithm.^[1] We employed *scikit-learn* library to establish ML model mentioned above, including support vector machine (SVM), neural network model (NNM), random forest (RF), k-nearest neighbor (KNN), naive Bayes (NB), and complex model.^[1]

Accuracy and precision, being two important model assessment criteria, are calculated by the following formulas:

$$\text{Accuracy} = \frac{TN+TP}{TN+TP+FP+FN} \quad (2)$$

$$\text{Precision} = \frac{TP}{TP+FP} \quad (3)$$

Where TP , FP , TN and FN represent the following: TP corresponds to samples of the positive category predicted as positive, FP corresponds to samples of the negative category predicted as positive, TN corresponds to samples of the negative category predicted as negative, and FN

corresponds to samples of the positive category predicted as negative. The model evaluation results were calculated using the RE-RCV method, which was newly proposed by us. All of the aforementioned processes as well as subsequent feature analysis were carried out using the PyCharm software.

Model evaluation method

Random-extracted and recoverable cross-validation method (RE-RCV): we extracted fixed-proportion data from established dataset as test data with calculating model accuracy once followed, after which, the extracted data were recycled to established dataset. Subsequently, the fixed-proportion test data were extracted from established dataset again, and that cycle was repeated 100 times. Finally, the average accuracy was calculated as ultimate model accuracy. Note here that it is guaranteed that the extracted test data should keep randomness.

Machine-learning code

```
import numpy as np
import random
import pandas as pd
from sklearn.preprocessing import MinMaxScaler, StandardScaler
from sklearn.ensemble import RandomForestClassifier, StackingClassifier
from sklearn.naive_bayes import GaussianNB
from sklearn.svm import LinearSVC
from sklearn.neural_network import MLPClassifier
from sklearn.neighbors import KNeighborsClassifier
```

```

data = pd.DataFrame(pd.read_excel("your_data.csv"))
num = list(range(len(data)))
test_size = int(0.2*len(num))
InputX = list(data)[a:b]
clf1 = LinearSVC(C=2, loss='squared_hinge', dual=False, tol=1e-4, multi_class=ovr)
clf2 = RandomForestClassifier(criterion="gini", n_estimators=400, max_depth=4,
                             max_features=sqrt)
clf3 = KNeighborsClassifier(n_neighbors=11, algorithm=auto, leaf_size=30,
                             weights=uniform, metric=minkowski)
clf4 = GaussianNB(priors=None, var_smoothing=1e-9)
clf5 = MLPClassifier(activation="relu", alpha=0.0001, early_stopping=False, shuffle=True,
                     hidden_layer_sizes=(64, 32, 16), learning_rate="constant",
                     learning_rate_init=0.01, solver="adam", validation_fraction=0.1,
                     max_iter=10000)
estimator1 = [clf1, clf2, clf3, clf4, clf5]
for n in estimator1:
    for m in estimator1:
        accuracy = []
        nums = random.sample(range(0, 1000), 100)
        for i in nums:
            test_data = data.sample(n= test_size, random_state=i)
            train_data = data.drop(test_data.index)
            x_test_data = test_data[InputX]
            x_train_data = train_data[InputX]
            y_test_data = test_data["Output Y"]
            y_train_data = train_data["Output Y"]
            transfer = StandardScaler()
            x_train = transfer.fit_transform(x_train_data)
            x_test = transfer.fit_transform(x_test_data)

```

```

estimator = StackingClassifier(estimators=[("estimator1", n),
("estimator2", m)])

estimator.fit(x_train, y_train_data)

score = estimator.score(x_test, y_test_data)

accuracy.append(score)

print(accuracy)

print("model accuracy: \n", np.mean(accuracy))

```

DFT calculation

These physicochemical traits containing Number, DM, Gap, Polar, R2, Zpe, and CV of all collected molecules (marked by orange in Table S1, Supporting Information) were obtained by first-principles calculations based on density functional theory (DFT).^[2,3] Molecular structure optimization and molecular surface electrostatic potential both are performed by Gauss09 at the M06-2X/def2TZVP level. Mapping of electrostatic potential the onto the molecular surface was performed with GaussView. Besides, the bonding between molecules and perovskite was performed via Vienna ab initio Simulation Package (VASP) based on DFT. The all-electron projected augmented wave (PAW) potential was used to describe the ion–electron interaction, and Perdew-BurkeErnzerhof (PBE) functional in the framework of generalized gradient approximation (GGA) was used to describe the exchange and correlation function.^[4] The detailed parameters are as followed during calculation: the energy cutoff of plane wave (520 eV), Energy convergence criteria (1×10^{-5} eV), and force convergence criteria (0.02 eV/Å). The DFT dispersion corrected (DFT+D3) method was used for describing the weak interaction.^[5] The interaction energies (E_{ads}) of different molecules with FAPbI₃ perovskite surface were calculated by $E_{\text{ads}} = E_{\text{total}} - E_{\text{pvk}} - E_{\text{m}}$, where E_{total} , E_{pvk}

and E_m are the total energies of the adsorption system, the FAPbI_3 slab model and different molecules, respectively. We used a vacuum of 20 Å along the z-direction for slab model.

Materials

Formamidinium iodide (FAI), methylammonium chloride (MACl), Cesium chloride (CsCl), and methylammonium iodide (MAI) were purchased from Greatcell Solar. Lead iodide (PbI_2) was purchased from TCI. Bis (trifluoromethylsulfonyl)-imide lithium salt (Li-TFSI) and 2,2',7,7'-tetrakis (N,N-di-p-methoxy-phenylamine)-9,9-spirobifluorene (Spiro-OMeTAD) were purchased from Xi'an Polymer Light Technology Corp. N,N-dimethylformamide (DMF), dimethyl sulfoxide (DMSO), chlorobenzene (CB), isopropanol (IPA), acetonitrile, 4-tertbutylpyridine (TBP), Phenolsulfonphthalein (Phen), 4-aminobenzenesulfonamide (4-A), 4-Amino-6-chlorobenzene-1 (4-A-6), 6-aminopyridine-3-carbonitrile (6-A-3), and Carzenide (Carz) were purchased from Sigma Aldrich, 4-Chloro-2-hydroxy-5-sulfamoylbenzoic acid (4-C) was purchased from Aladdin. All the chemicals were used as received without further purification.

Preparation of perovskite precursor solution

FAI (228.72 mg), MAI (11.13 mg), PbI_2 (709.96 mg), CsCl (11.79 mg) and MACl (28.36 mg) were dissolving into 1 mL of mixture solvent of DMF/DMSO (4:1 volume ratio) as the perovskite precursor solution. For molecular passivation, the perovskite precursors were obtained by adding different molecules (4-A: 1.0 mg/mL, 4-C: 1.0 mg/mL, Phen: 1.0 mg/mL, 4-A-6: 0.5 mg/mL, 6-A-3: 1.0 mg/mL, or Carz: 1.0 mg/mL) into precursor. The

mixed solution was continuously stirring for 12 h and then was used without filtration.

Device fabrication. Fluorine-doped tin oxide (FTO)/glass substrate was cleaned by ultra-sonication in diluted detergent solution (detergent mixed with deionized water in volume ratio of 1:100), followed by subsequent washes with acetone, ethyl alcohol and deionized water, each step lasting for 20 min. After 30 min of ultraviolet (UV)-O₃ treatment, a multiple electron-transporter-layer combined with TiO₂ and SnO₂ was deposited as previous reported process.^[6] Then the pre-prepared perovskite precursor solution was spin-coated onto FTO/TiO₂/SnO₂ substrate at 2500 rpm for 10s, followed with drying in a gas pump system under a constant pressure of 10 Pa for 1 min. Subsequently, the sample was immediately annealed at 150 °C for 20 min. After cooling, the Spiro-OMeTAD solution (72.3 mg of Spiro-OMeTAD, 30.6 μL of tBP, and 19.5 μL of LiTFSI (520 mg/mL in acetonitrile) in 1 mL chlorobenzene) was spin coated onto top of the perovskite film at 3000 rpm for 30 s followed by an 8-hour oxidation period in dry environment. Finally, a 80 nm of gold (Au) electrode was deposited by thermal evaporation. An antireflective film (NANOECOWAY Co. Ltd.) was applied to glass side of PSCs.

Characterization and measurements

The *J-V* curves of the PSCs were obtained by using a 2400 series source meter (Keithley Instruments) under illumination of a solar simulator (Newport, Class AAA) with AM 1.5G filter (Sol3A, Oriel) at light intensity of 100 mW cm⁻², calibrated by a standard Si reference cell. The active area of PSCs was determined by metal mask with aperture of 0.06

cm². The PL spectra were obtained using a super-resolution confocal microscope (Leica TCS SP8 STED 3X). The XPS measurements were performed with a PHI 5600 ultrahigh vacuum system using a hemispherical electron energy analyzer and an Al K α source (1486.6 eV, PHI 04-548 dual anode X-ray source) for excitation. ¹H NMR spectra were collected using AVANCE III HD 600 MHz instrument spectrometer. The incident photon-to-current conversion (IPCE) values were confirmed as a function of wavelength from 300 to 900 nm (IQE200B, Oriel) for devices without bias light.

Statistical Analysis

Before training the model, it is essential to normalize all input X data. This normalization process is accomplished using the StandardScaler algorithm. For statistical analysis of perovskite solar cell data, we prepared 14 cells under the same parameters to conduct statistical analysis. In addition, we introduced the coefficient of variation to assess the dispersion of two model evaluation methods (RE-RCV and K-Fold). We calculated the model accuracy 50 times using each method, then obtained the coefficient of variation (CV) using the following formula:

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}} \quad (1)$$

$$CV = \frac{s}{\bar{x}} \times 100\% \quad (2)$$

where s and $\bar{x}$ are standard deviation and average value, respectively. A smaller coefficient of variation indicates lower dispersion.”

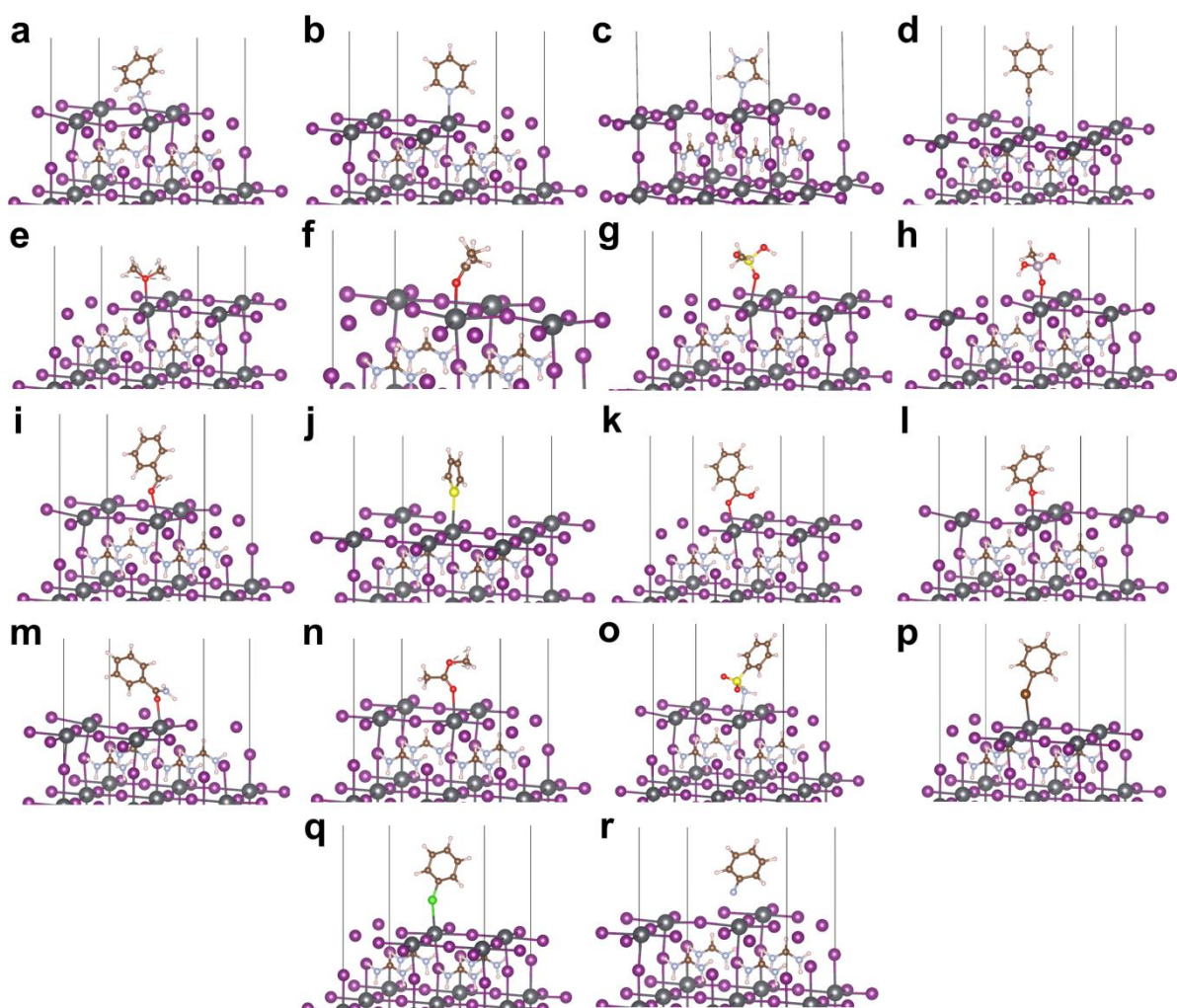


Figure S1. Bonding conditions between the Lewis acid/base groups of molecules and uncoordinated Pb^{2+} ions of perovskite: a) Amino, b) Pyridine, c) Imidazole, d) Cyano group, e) Ether group, f) Carbonyl group, g) Sulfonic acid group, h) Phosphate group, i) Aldehyde group, j) Thiophene, k) Carboxylic acid group, l) Hydroxy, m) Acetyl, n) Ester group, o) Sulfonyl group, p) Br, q) Cl, r) F. It's worth noting that the r) F group is unable to bind with uncoordinated Pb^{2+} ions.

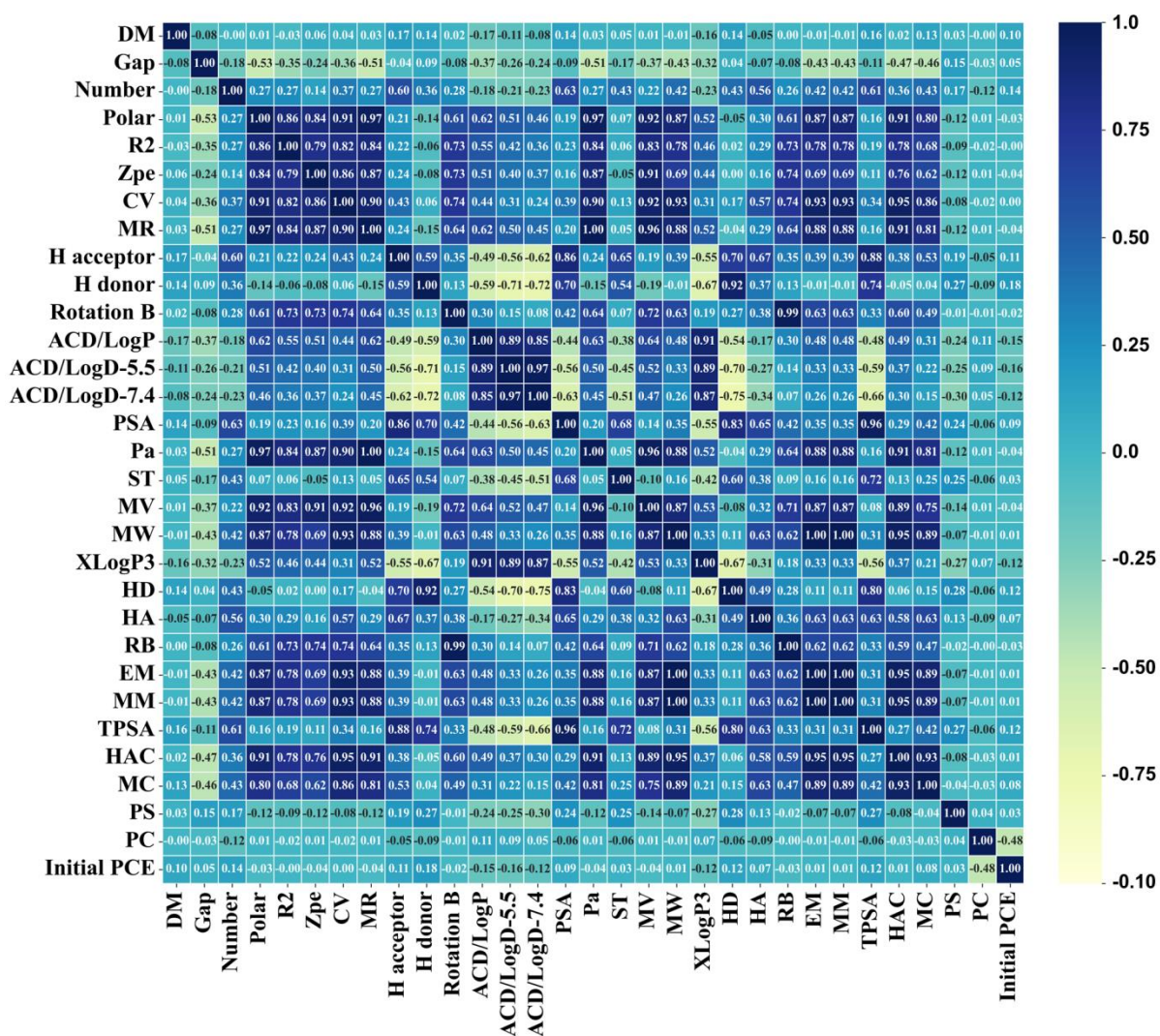


Figure S2. Correlation matrix of Pearson coefficients among 31 untreated features.

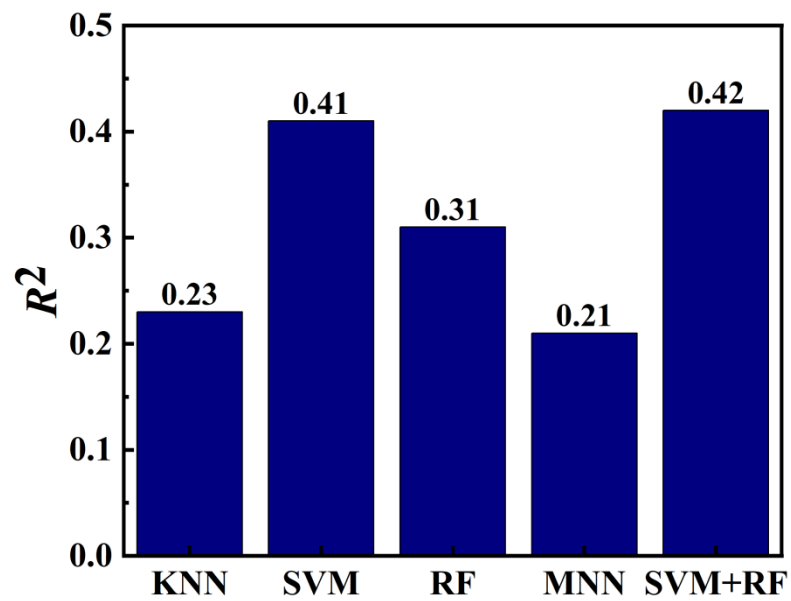


Figure S3. The coefficient of determination (R^2) of single ML model and complex ML models based on evaluation method of RE-RCV.

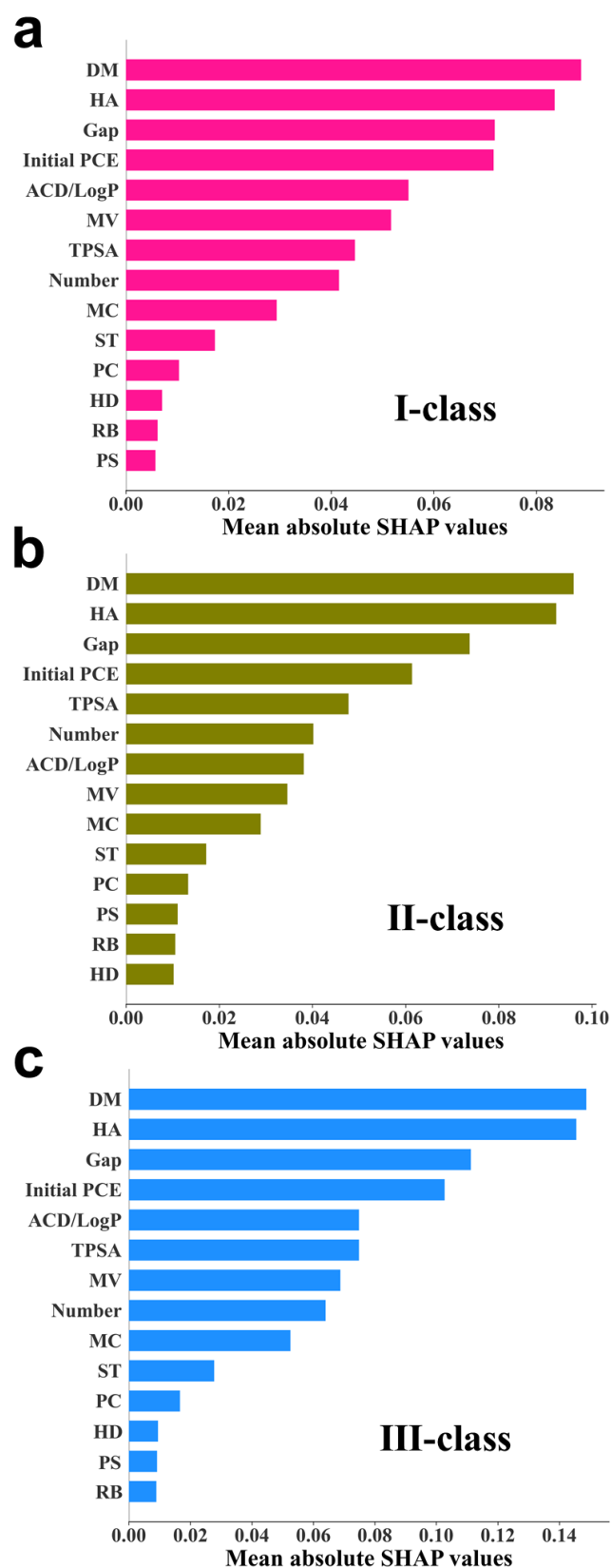


Figure S4. SHAP values rankings of Input X for different Output class: a) I-class, b) II-class, and c) III-class.

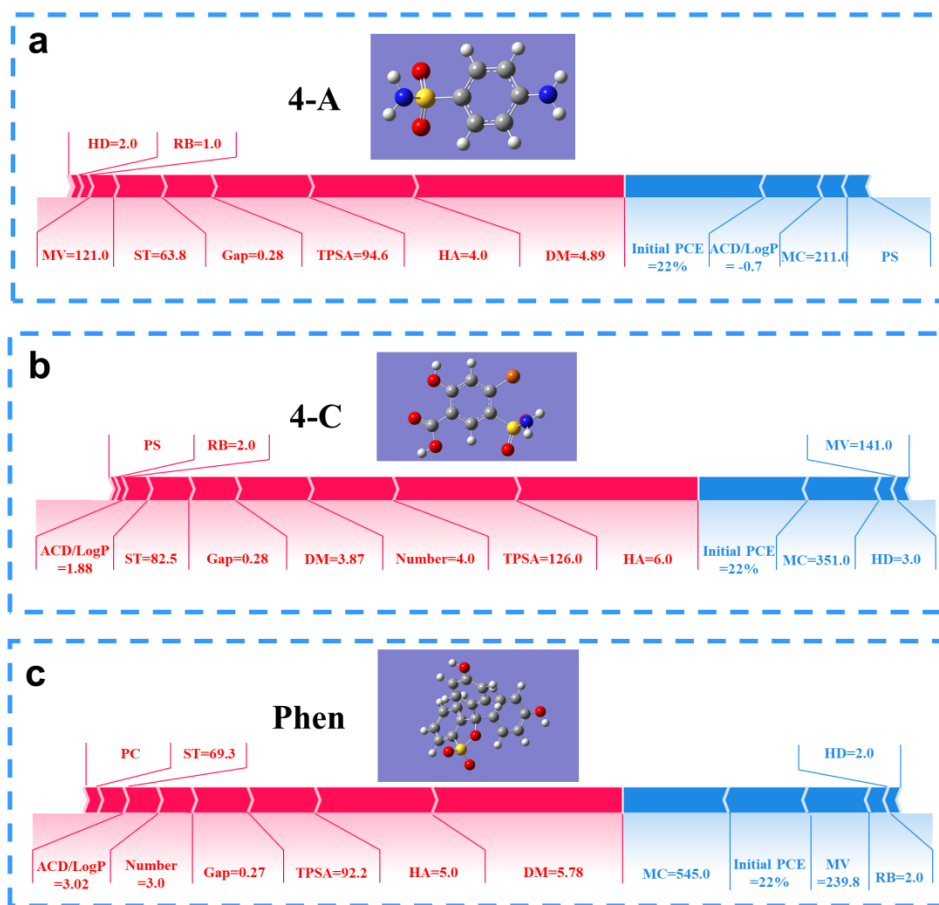


Figure S5. The detailed contribution degree related to SHAP values of molecular traits to the probability of classification prediction: a) 4-A, b) 4-C and c) Phen.

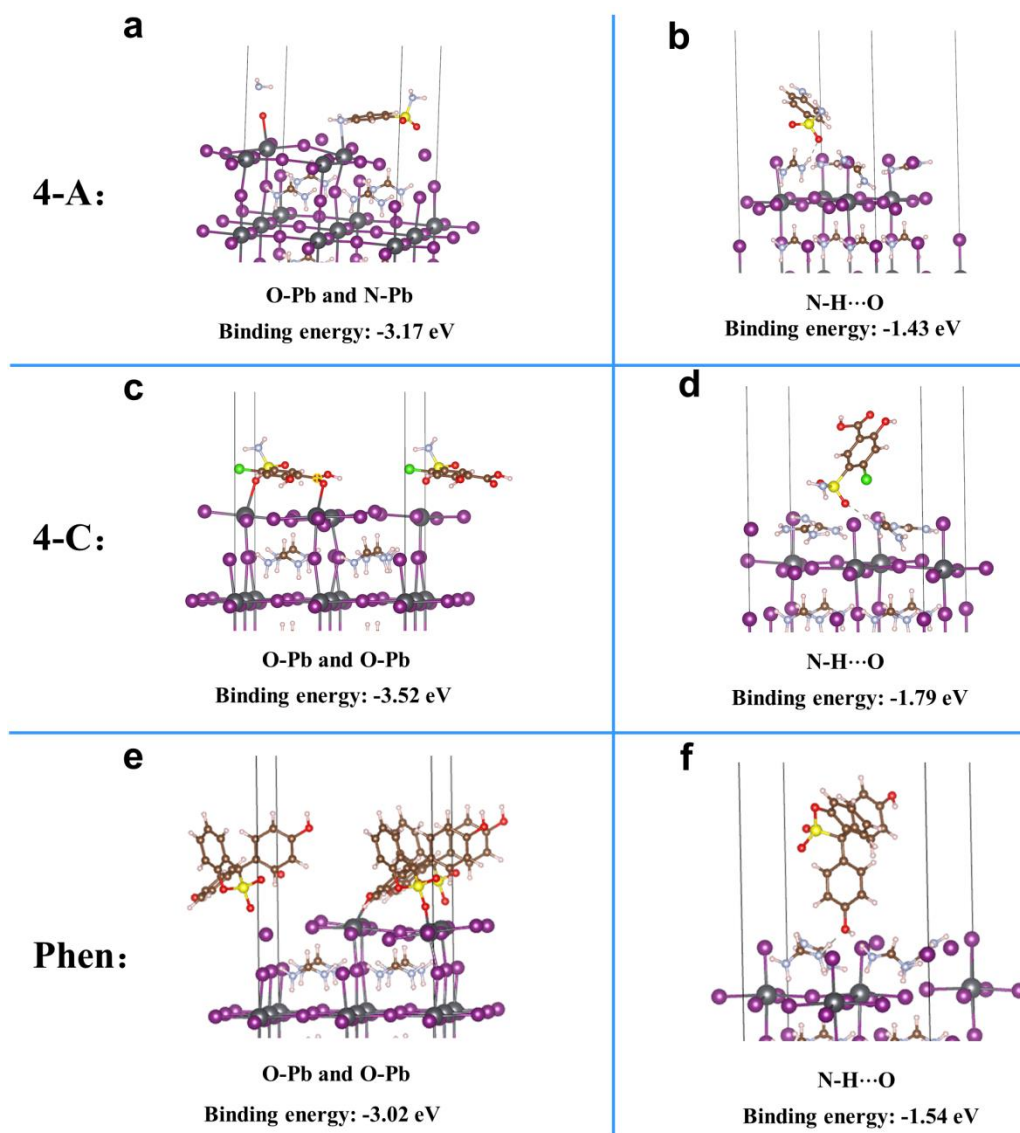


Figure S6. Binding energies for the interactions of a, b) $\text{NH}_2\text{-Pb}$, O-Pb, and N-H...O for 4-A; c, d) O-Pb and N-H...O for 4-C; e, f) O-Pb and N-H...O for Phen.

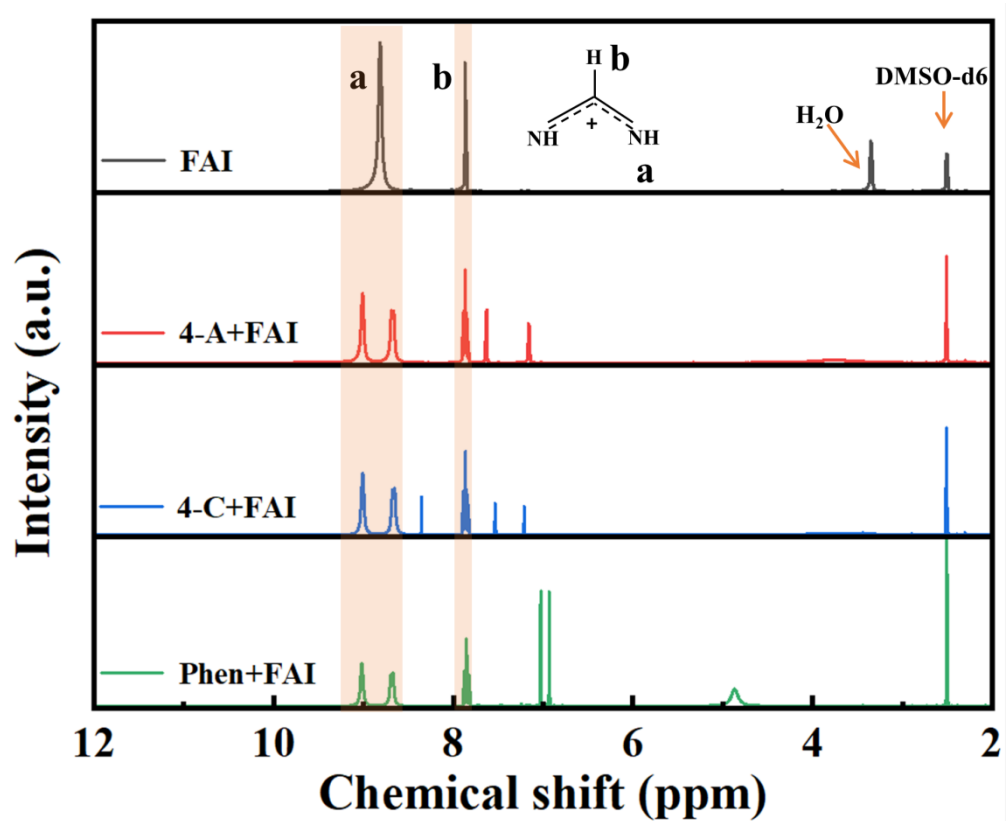


Figure S7. A comparison of ^1H NMR spectra of FAI, 4-A+FAI, 4-C+FAI, and Phen+FAI.

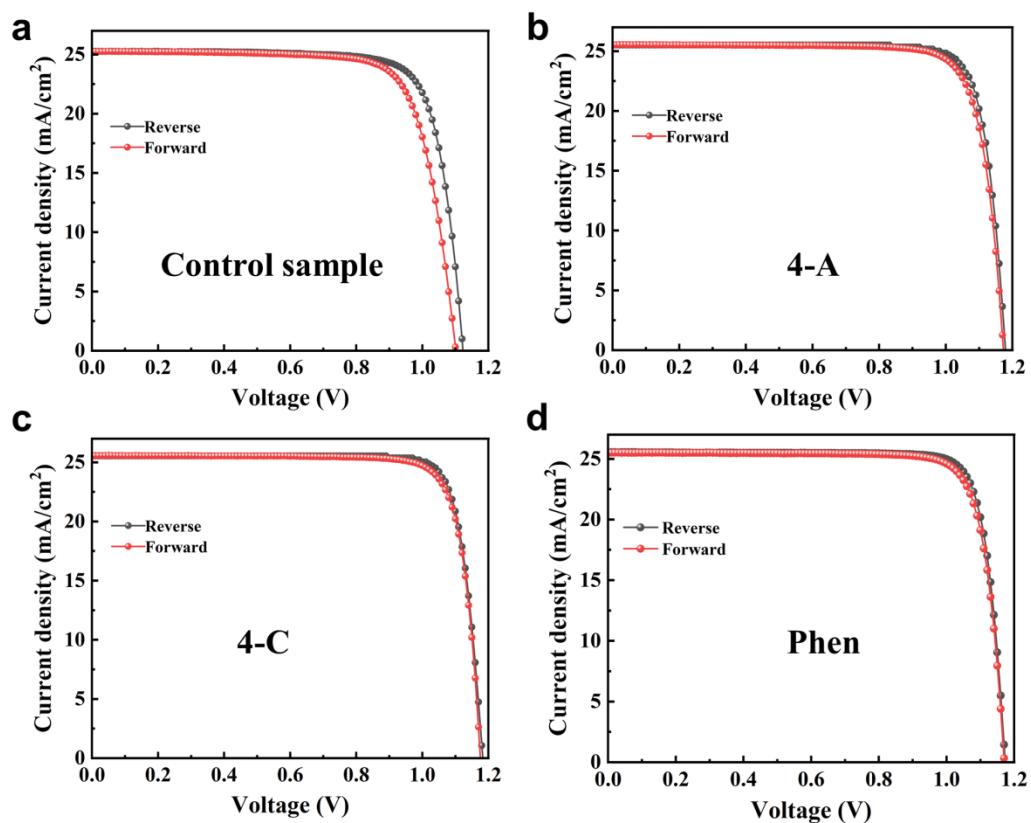


Figure S8. *J*-*V* curves under reverse (1.2 V to 0 V) and forward (0 V to 1.2 V) scan of the champion CsFAMA-based PSCs a) without passivation treatment and with b) 4-A, c) 4-C, and d) Phen passivation treatment, respectively.

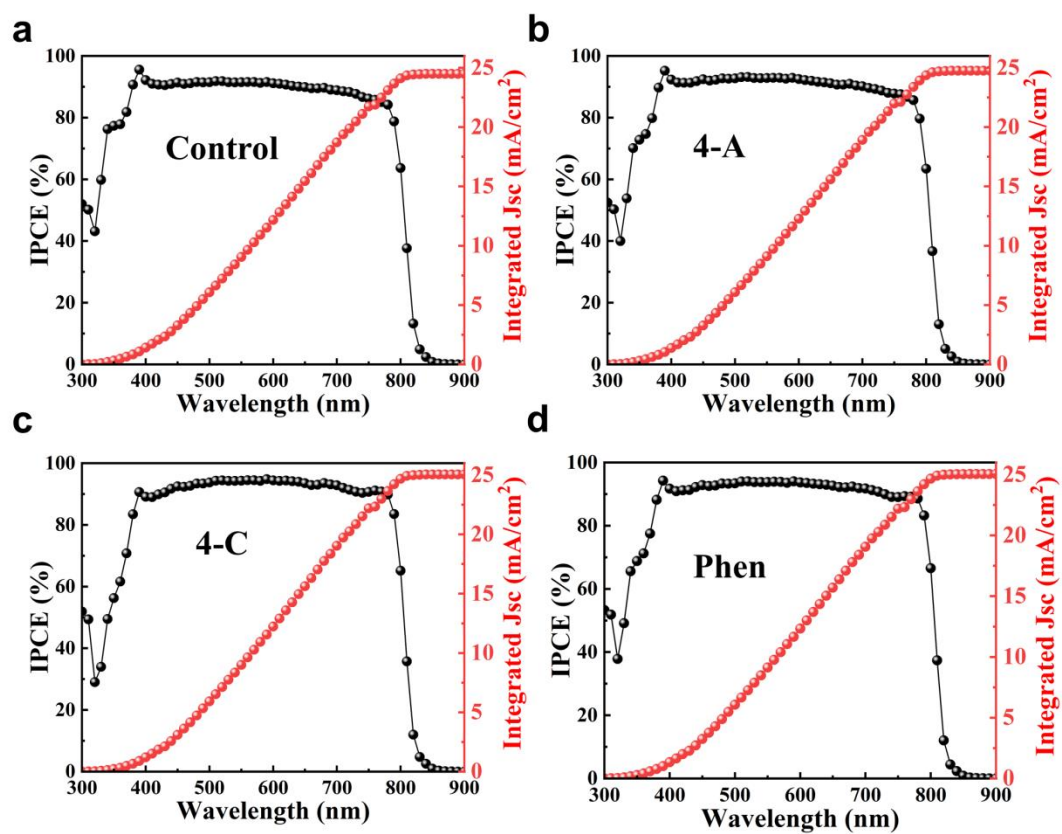


Figure S9. The IPCE and the integrated J_{sc} of the CsFAMA-based PSCs a) without passivation treatment and with b) 4-A, c) 4-C, and d) Phen passivation treatment, respectively.

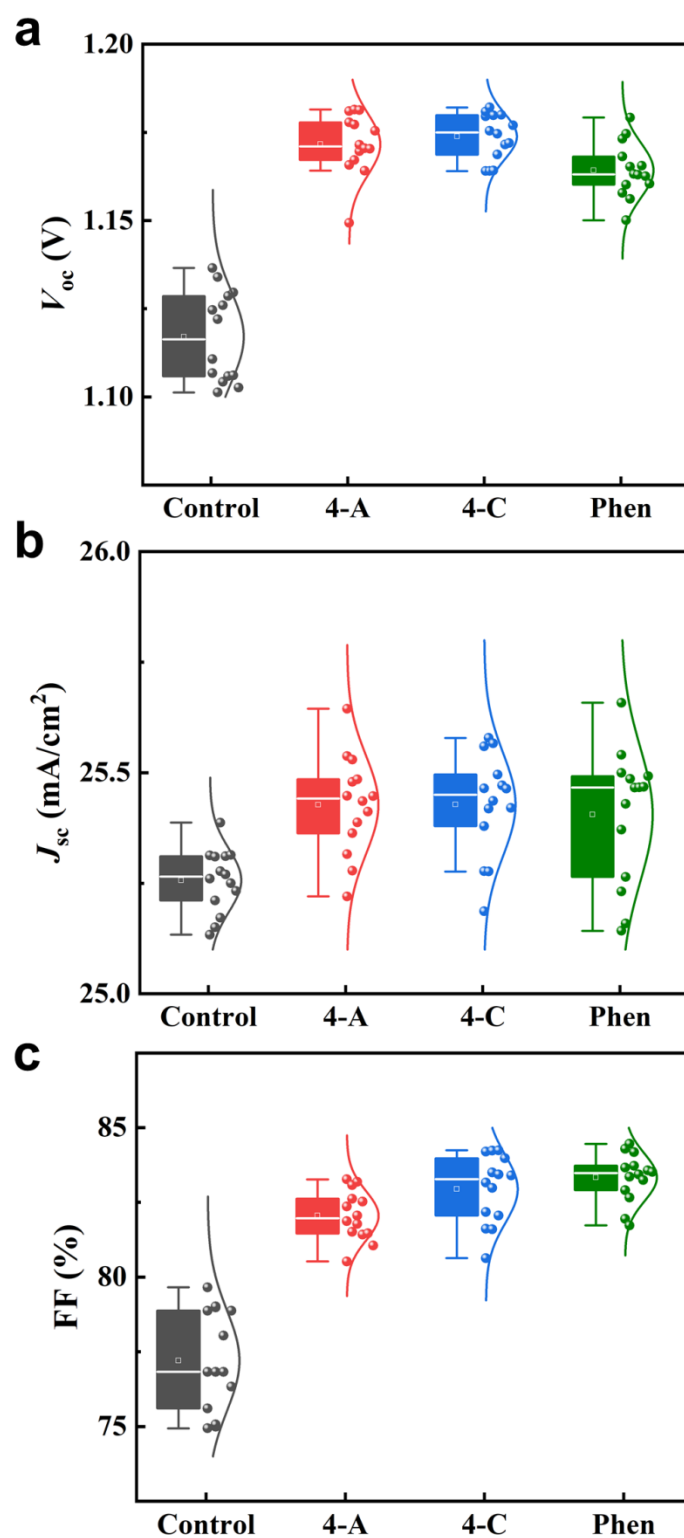


Figure S10. Statistics of a) V_{oc} , b) J_{sc} and c) FF of PSCs before and after passivation treatment by 4-A, 4-C or Phen.

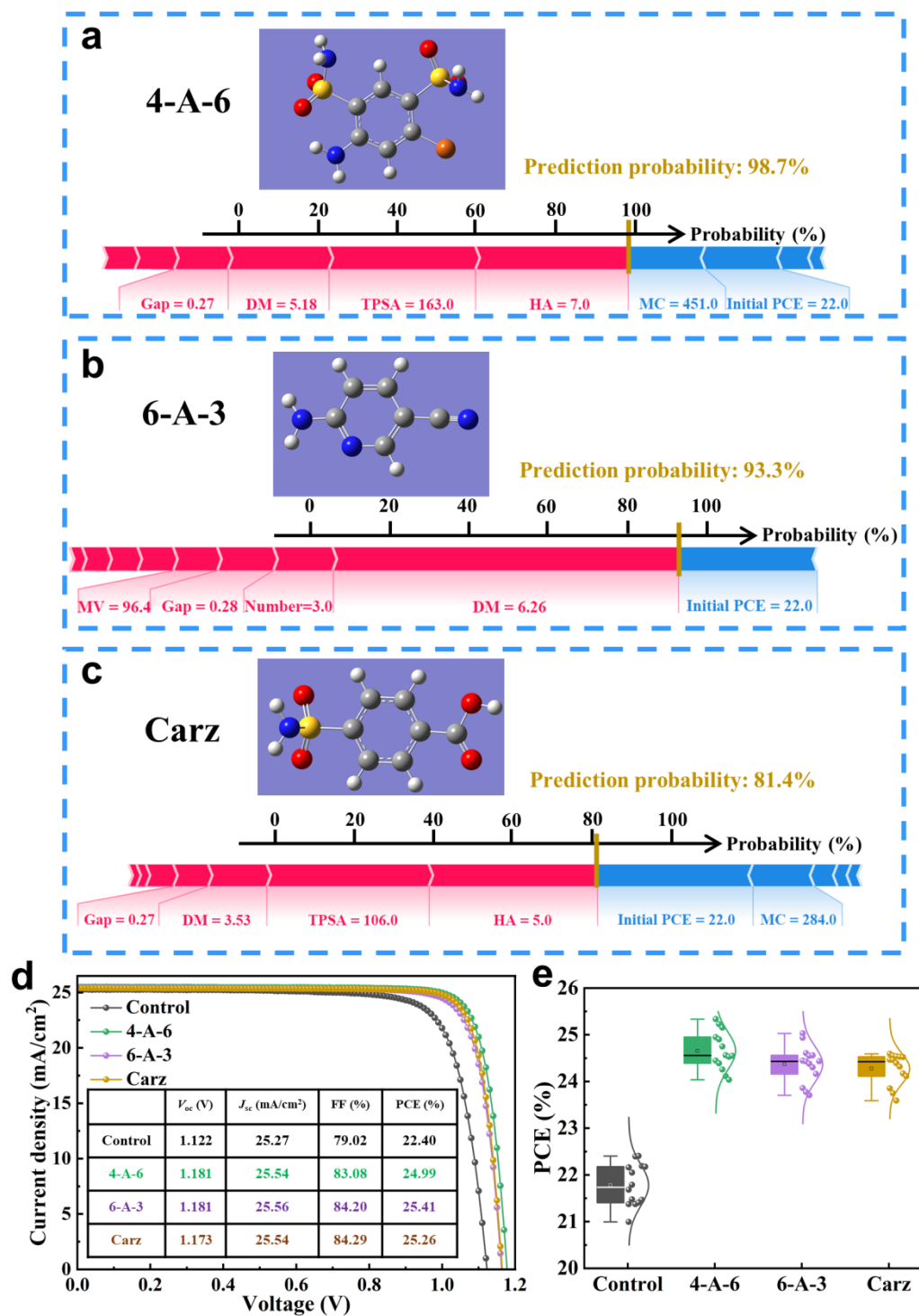


Figure S11. Configurations and predicted probabilities of screened small molecules by S-R ML model: a) 4-A-6, b) 6-A-3 and c) Carz. The length of the color bar corresponds to the contribution degree related to SHAP values of dominant molecular traits to the probability of classification prediction, with red and blue denoting positive and negative roles, respectively.

The intersection between red and blue zones corresponds to the prediction probability of classifying that molecule as III-class. d) *J-V* curves and e) statistical PCE of CsFAMA-based PSCs with and without 4-A-6, 6-A-3 and Carz treatment.

Table S1. Definitions and detailed interpretations of 31 features. The feature descriptions of molecular traits are color-coded: features in orange are obtained by DFT calculation, features in blue are obtained from ChemSpider, and features in green are obtained from PubChem. Features in black are perovskite film properties and passivation style.

Feature	Feature definition	Detailed interpretation
Number	Number of coordination group	The number of coordination group whether the Lewis acid/base groups of molecule bind with uncoordinated Pb^{2+} by DFT calculation
DM	Dipole moment	the net or average separation of charge in a molecule
Gap	HOMO-LUMO Gap	the energy difference between a molecule's Highest energy Occupied Molecular Orbital (HOMO) and its Lowest energy Unoccupied Molecular Orbital (LUMO)
Polar	Polarizability	the capability of molecular electronic system to be distorted by the external field

R2	Electronic spatial extent	a measure of molecular volume
Zpe	Zero point vibrational energy	The energy difference between the lowest point on the potential energy surface (equilibrium energy) and the energy of the vibrationless energy level ($v=0$).
CV	Heat capacity	the amount of heat required to raise 1 mole of a substance by 1 degree Kelvin.
MR	Molar refractivity	Measure of the total polarizability of a mole of a substance and is dependent on the temperature, the index of refraction, and the pressure
H acceptor	H bond acceptors	The number of H bond acceptors in molecule
H donor	H bond donors	The number of H bond donors in molecule
Rotation B	Freely Rotating Bonds	The number of freely rotating bonds in molecule
ACD/LogP	Lipid-water partition coefficient	Lipid–water partition coefficient is used to characterize the molecular hydrophobicity
ACD/LogD-5.5	octanol-water distribution coefficient at pH5.5	the ratio of the equilibrium concentrations of the chemical in the two phases at pH 5.5
ACD/LogD-7.4	octanol-water distribution coefficient at pH 7.4	the ratio of the equilibrium concentrations of the chemical in the two phases at pH 7.4

PSA	Polar surface area	the combined surface area belonging to oxygen and nitrogen atoms and hydrogen atoms bound to these electronegative atoms
Pa	Polarizability	the capability of its electronic system to be distorted by the external field
ST	Surface tension	the tendency of fluid surfaces to shrink into the minimum surface area possible
MV	Molar volume	the volume of one mole of molecule at a specified pressure and temperature
MW	Molecular weight	mass of a molecule of a substance, based on 12 as the atomic weight of carbon-12
XLogP3	Lipid–water partition coefficient	Lipid–water partition coefficient is used to characterize the molecular hydrophobicity
HA	Hydrogen bond acceptor count	The number of hydrogen bond acceptor in molecule
HD	Hydrogen bond donor count	The number of hydrogen bond donor in molecule
RB	Rotatable bond count	The number of rotatable bond in molecule
EM	Exact Mass	Exact Mass is calculated by adding the exact masses of the most abundant isotopes of the constituent elements.
MM	Monoisotopic mass	one of several types of molecular masses used in mass spectrometry

TPSA	Topological polar surface area	the sum of surfaces of polar atoms in a molecule
HAC	Heavy Atom Count	Heavy Atom Count is the total number of non-hydrogen atom ('heavy' atom) within a molecule
MC	Molecular complexity	Molecular complexity is calculated by counting the set bits in a molecular fingerprint
PS	Passivation style	bulk passivation, top surface passivation and bottom surface passivation
PC	Perovskite composition	the changes of X-site anion (I^- and Br^-) and A-site cation (MA^+ , FA^+ , and mixed)
Initial PCE	PCE of PSCs without treatment	the un-passivated film quality

Table S2. Detailed hyperparameters of the single ML classification algorithms used in this study.

ML classification algorithms	Hyperparameters in scikit-learn
K-nearest neighbor (KNN)	n_neighbors: 11; algorithm: auto; leaf_size: 30; weights: uniform; metric: minkowski
Naive Bayes (NB)	priors: None; var_smoothing: 1e-9

Support vector machine (SVM)	C: 4; loss: squared_hinge; Dual: False; tol: 1e-4; multi_class: ovr
Random forest (RF)	Criterion: gini; n_estimators: 400; Max_depth: 4; max_features: sqrt
Neural network model (NNM)	activation: relu; alpha: 0.0001; early_stopping: False; shuffle: True; hidden_layer_size: (64, 32, 16); learning_rate: constant; slover: adam; max_iter: 10000; validation_fraction: 0.1

Table S3. Feature values of key molecular traits for the three screened molecules.

Feature	4-aminobenzene -sulfonamide (4-A)	4-Chloro-2-hydroxy- 5-sulfamoylbenzoic acid (4-C)	Phenolsulfonphthalein (Phen)
DM (Debey)	4.892	3.867	4.363
Gap (eV)	0.283	0.275	0.21
Number	2	4	4
RB	1	2	1
TPSA ($\text{\AA}^2$)	94.6	126	92.5
ST (dyne/cm)	63.8	82.5	73.3
MV (cm^3)	121	141	178

MC	211	351	437
ACD/LogP	-0.7	1.88	-1.4
HA	4	6	4
HD	2	3	2

Table S4. Statistical photovoltaic parameters of CsFAMA-based PSCs and champion PSCs with and without 4-A, 4-C, Phen, 4-A-6, 6-A-3 or Carz treatment.

		V_{oc} (V)	J_{sc} (mA/cm ²)	FF (%)	PCE (%)
Control	Champion	1.137	25.27	79.02	22.40
	Average	1.119±0.018	25.26±0.13	77.30±2.36	21.70±0.70
With 4-A	Champion	1.181	25.54	83.08	24.99
	Average	1.165±0.016	25.43±0.21	81.89±1.37	24.36±0.63
With 4-C	Champion	1.182	25.56	84.20	25.41
	Average	1.173±0.009	25.38±0.20	82.44±1.80	24.76±0.65
With Phen	Champion	1.173	25.54	84.29	25.26
	Average	1.165±0.015	25.40±0.26	83.09±1.37	24.56±0.70

With 4-A-6	Champion	1.178	25.50	84.36	25.33
	Average	1.165±0.015	25.40±0.13	82.55±1.81	24.68±0.65
With 6-A-3	Champion	1.164	25.39	84.66	25.03
	Average	1.166±0.014	25.38±0.22	82.62±2.04	24.37±0.66
With Carz	Champion	1.163	25.45	83.07	24.59
	Average	1.165±0.014	25.33±0.12	82.40±1.56	24.09±0.50

Table S5. A statistical comparison of the price and passivation effect of different passivation molecules.

Reference	Molecule	Price (\$/g)	Best PCE (%)	ΔPCE (%)
[7]	4-(N,N-Diphenylamino) benzaldehyde	36.2	23.20	2.43
[7]	4-Phenylacetophenone	2.1	22.47	1.70
[7]	4-Phenylbenzaldehyde	8.7	22.85	2.08
[8]	1H-Benzimidazole	15.2	24.28	1.56
[9]	L-Theanine	1186.0	24.58	2.29
[10]	Sulphanilic acid	12.4	24.09	2.60
[11]	1,3,3-Trimethylindolino-6'-nitrobenzopyrrolospiran	150.0	23.23	1.27
[12]	Sulfamide	7.2	24.64	2.34
[13]	1,3,5-Tri(thiophen-2-yl)benzene	195.8	22.46	2.90

[14]	2-Pyridyltriflimide	50.2	22.90	2.16
This work	4-Chloro-2-hydroxy-5-sulfamoylbenzoic acid (4-C)	10.3	25.41	3.01
	4-aminobenzenesulfonamide (4-A)	1.0	24.99	2.59
	Phenolsulfonphthalein (Phen)	5.9	25.26	2.86
	4-Amino-6-chlorobenzene-1 (4-A-6)	5.6	25.33	2.93
	6-aminopyridine-3-carbonitrile (6-A-3)	6.1	25.03	2.63
	Carzenide (Carz)	7.4	24.59	2.19

References

- [1] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, *J. Mach. Learn. Res.* **2011**, *12*, 2825.
- [2] G. Kresse, J. Hafner, *Phys. Rev. B* **1993**, *48*, 13115.
- [3] G. Kresse, J. Furthmüller, *Phys. Rev. B* **1996**, *54*, 11169.
- [4] J. P. Perdew, K. Burke, M. Ernzerhof, *Phys. Rev. Lett.* **1996**, *77*, 3865.
- [5] K. Lee, É. D. Murray, L. Kong, B. I. Lundqvist, D. C. Langreth, *Phys. Rev. B* **2010**, *82*, 081101.
- [6] B. Ding, S.-Y. Huang, Q.-Q. Chu, Y. Li, C.-X. Li, C.-J. Li, G.-J. Yang, *J. Mater. Chem. A* **2018**, *6*, 10233.
- [7] X. Jiang, B. Zhang, G. Yang, Z. Zhou, X. Guo, F. Zhang, S. Yu, S. Liu, S. Pang, *Angew. Chem. Int. Ed.* **2023**, *62*, e202302462.
- [8] W. Zhang, L. He, Y. Meng, H. Kanda, D. Tang, B. Ding, Y. Ding, M. K. Nazeeruddin, X. Li, *Adv. Energy Mater.* **2022**, *12*.

- [9] Y. Li, L. Liu, C. Zheng, Z. Liu, L. Chen, N. Yuan, J. Ding, D. Wang, S. Liu, *Adv. Energy Mater.* **2023**, *13*, 2203190.
- [10] K. I. Wang, M. Li, Y. H. Lou, J. Chen, Y. R. Shi, C. H. Chen, Y. H. Zhou, Z. K. Wang, L. S. Liao, *Adv. Energy Mater.* **2023**, *13*, 2203471.
- [11] Y. R. Shi, K. L. Wang, Y. H. Lou, G. L. Liu, C. H. Chen, J. Chen, L. Zhang, Z. K. Wang, *Adv. Mater.* **2022**, *34*, e2205338.
- [12] Y. Wu, B. Chang, L. Wang, H. Li, L. Pan, Z. Liu, L. Yin, *Adv. Mater.* **2023**, *35*, 2300174.
- [13] R. Zhuang, L. Wang, J. Qiu, L. Xie, X. Miao, X. Zhang, Y. Hua, *Chem. Eng. J.* **2023**, *463*, 142449.
- [14] C. Tian, Y. Zhao, X. Han, B. Li, Y. Rui, H. Xiong, Y. Qiu, W. An, K. Li, C. Hou, Y. Li, H. Wang, Q. Zhang, *Chem. Eng. J.* **2023**, *452*, 139345.